

MARKER SELECTION

Learned gene-gene graph



Expression



$$x \in \mathbb{R}^N$$

(N genes)

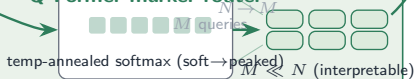
smooth x (denoise)

Smoothed expression

$$\tilde{x} = \hat{G} x$$

$\hat{G}$ — low-rank, end-to-end

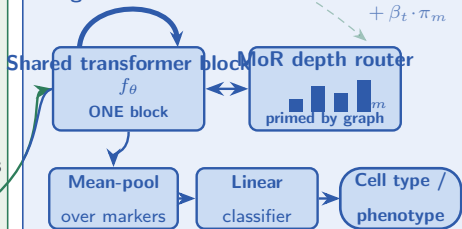
Q-Former marker router



Marker tokens

RECURSIVE ROUTING & CLASSIFICATION

weight-shared $\times K$



Gene-graph ablation — single-cell macro-F1
honest negative

